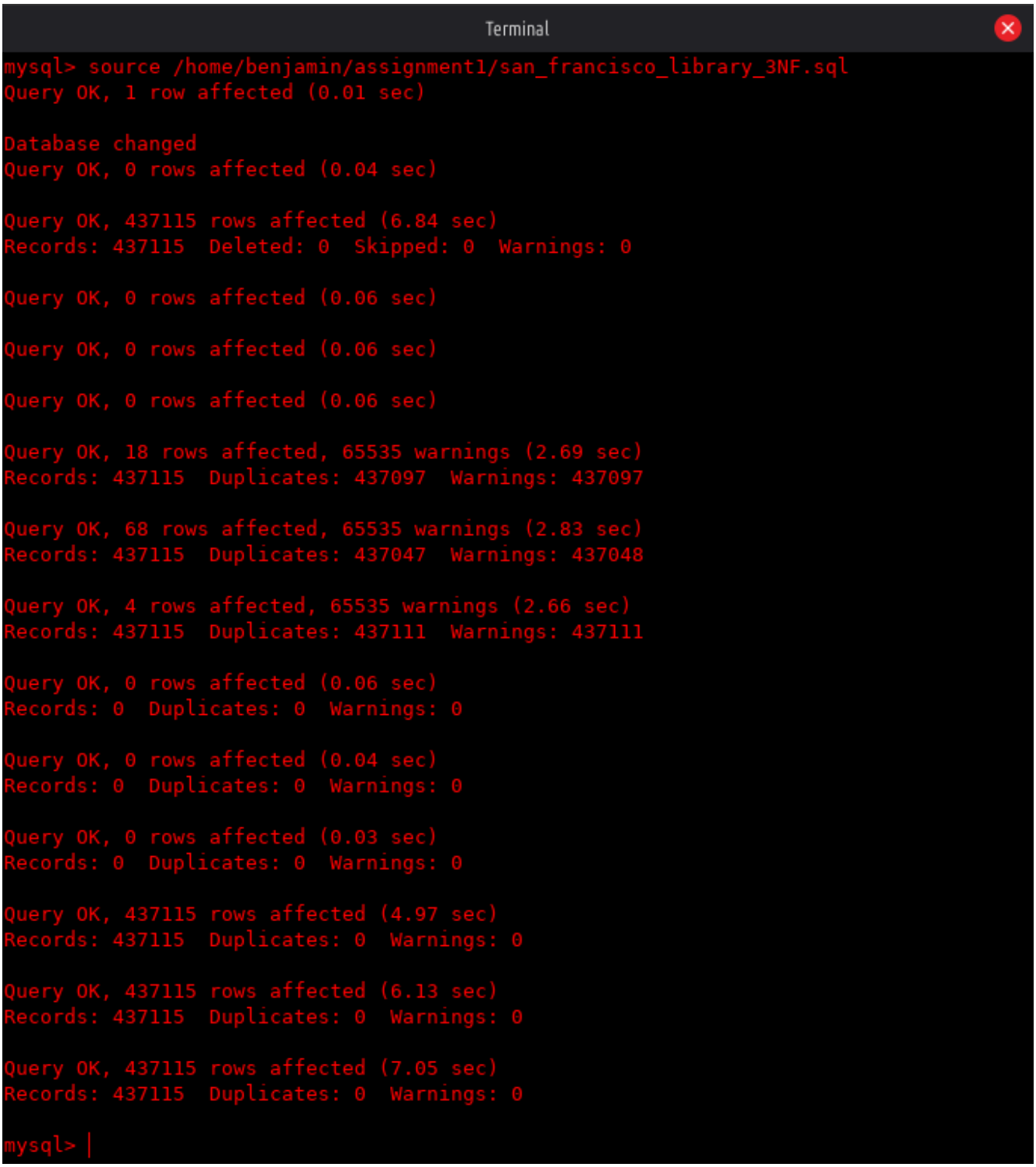
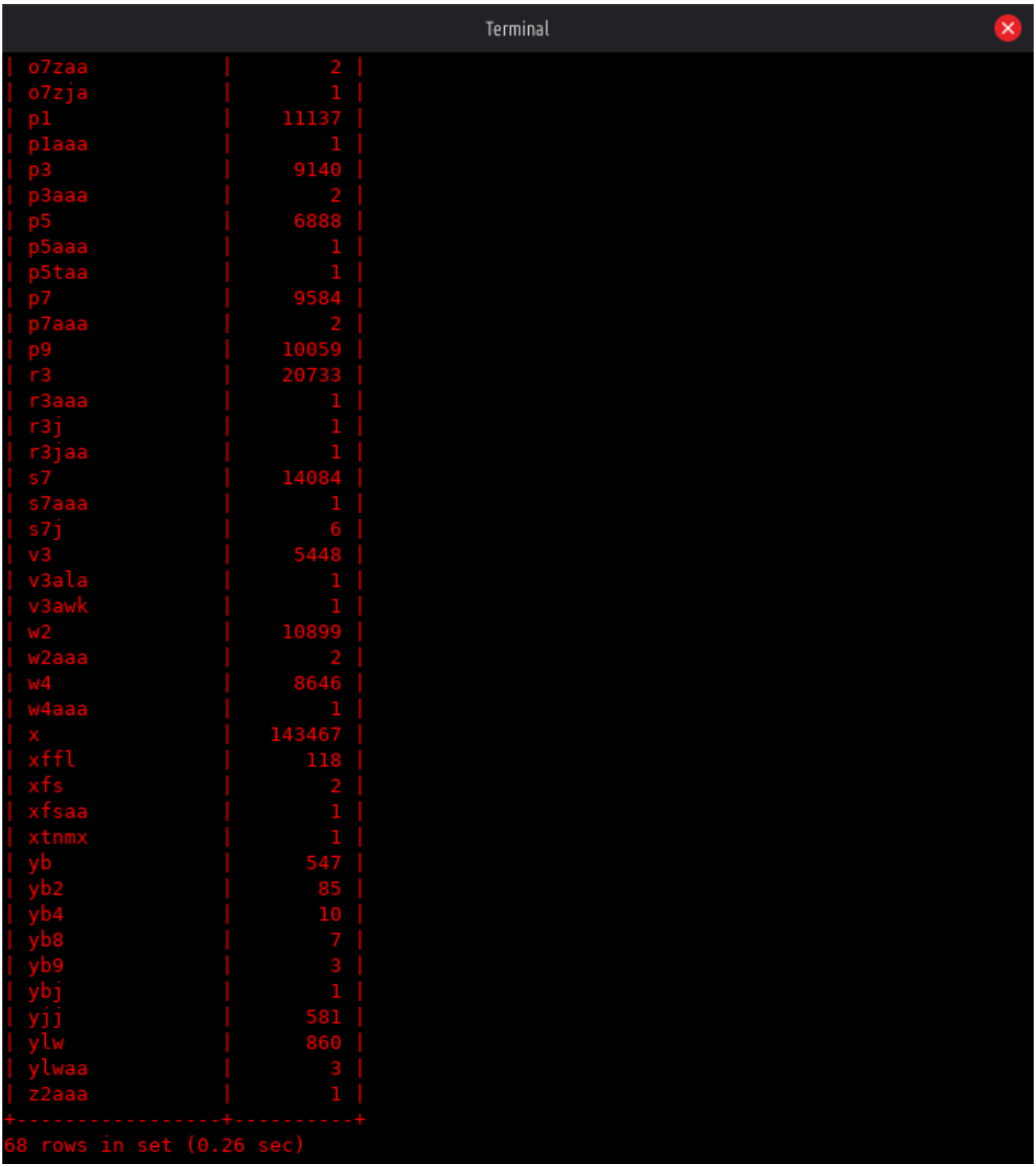
It takes slightly over half a minute to build the san_francisco_library_3NF database from scratch.



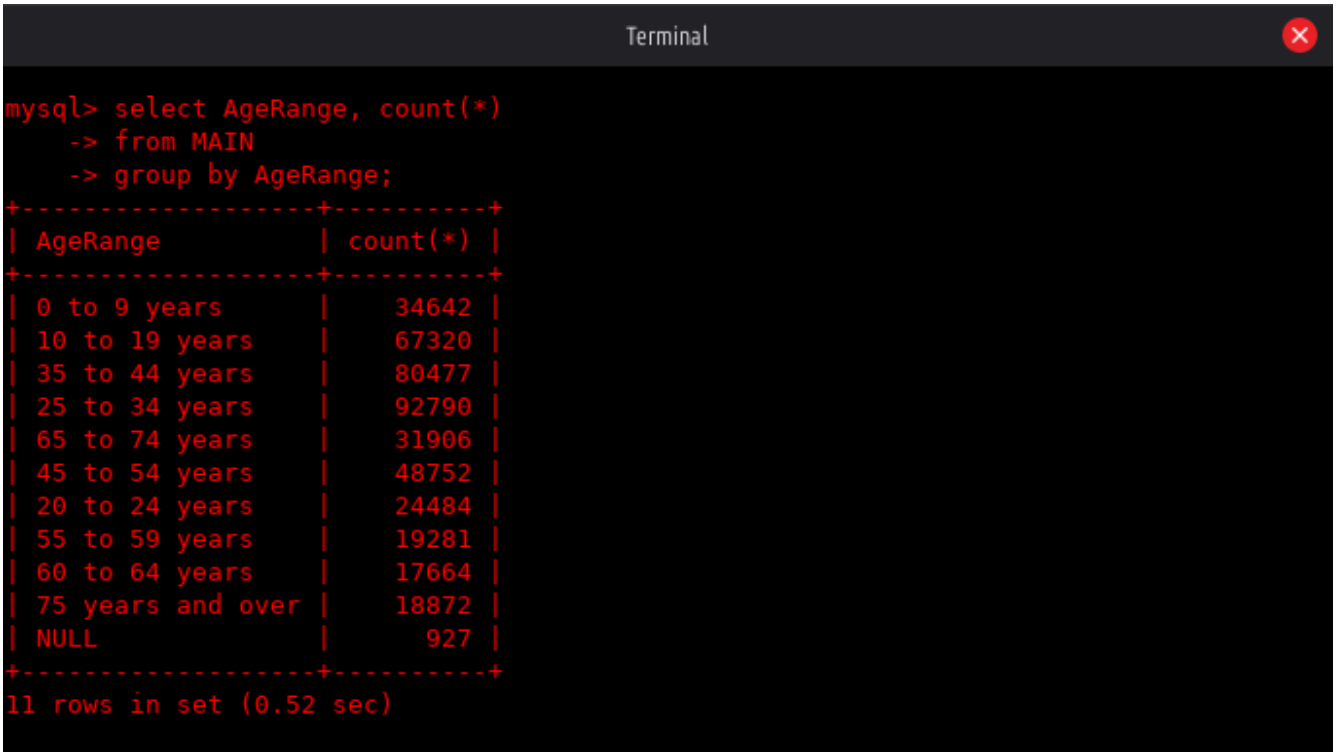
My first finding is that the data is an absolute mess. There are a ton of typos, mismatched Nones, Unknowns, and NULLS all mixed in together, disparate types of data sharing the same column (Patron Type Code), and so on. This makes any kind of analysis difficult, but I will do what I can. I will also include performance metrics in each screenshot.

The only home library with over 100,000 visits is the main library.

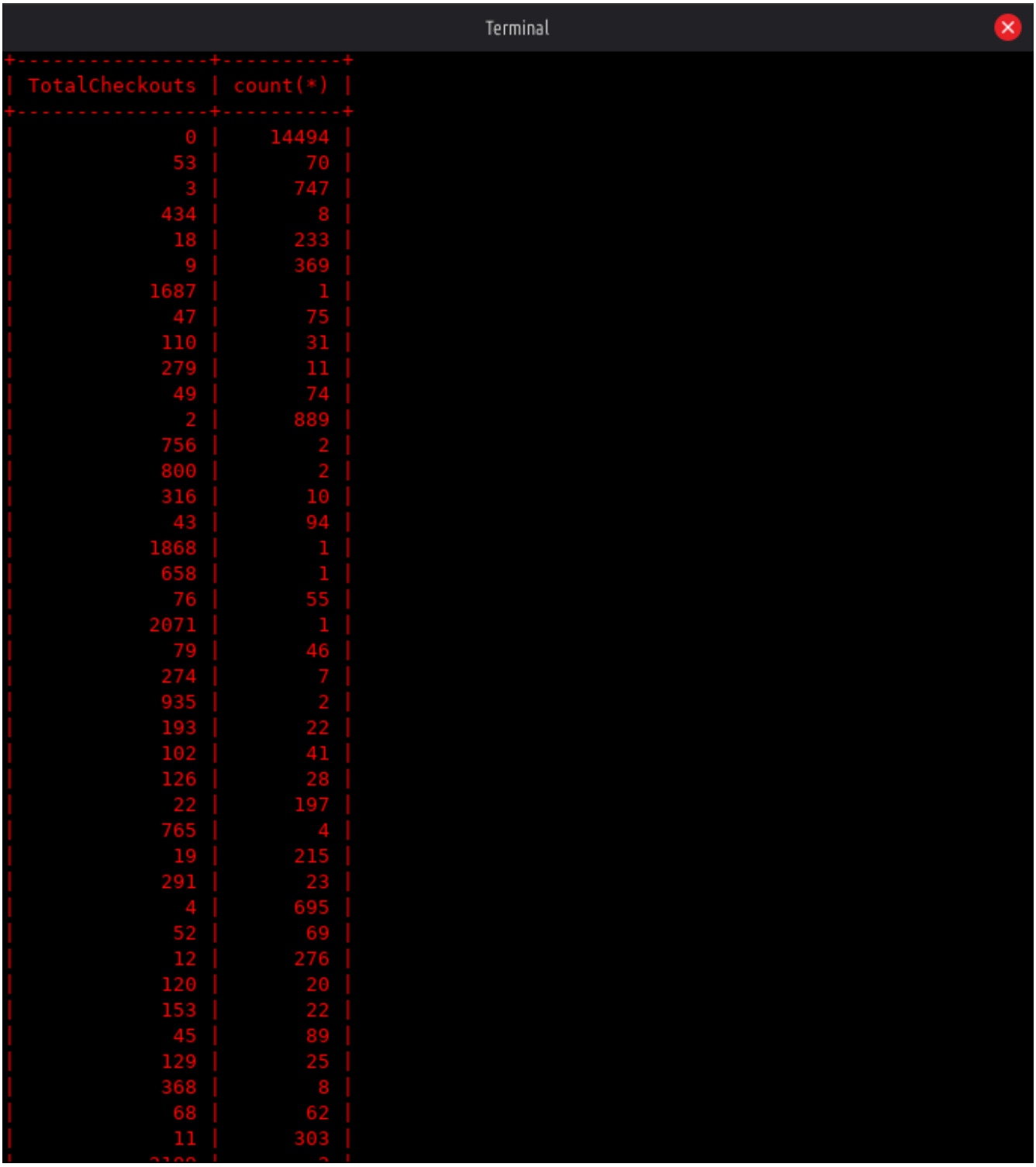


This was found using “select HomeLibraryCode, count(*) from MAIN group by HomeLibraryCode;”.

The most common age range appears to be 25 to 34 years.



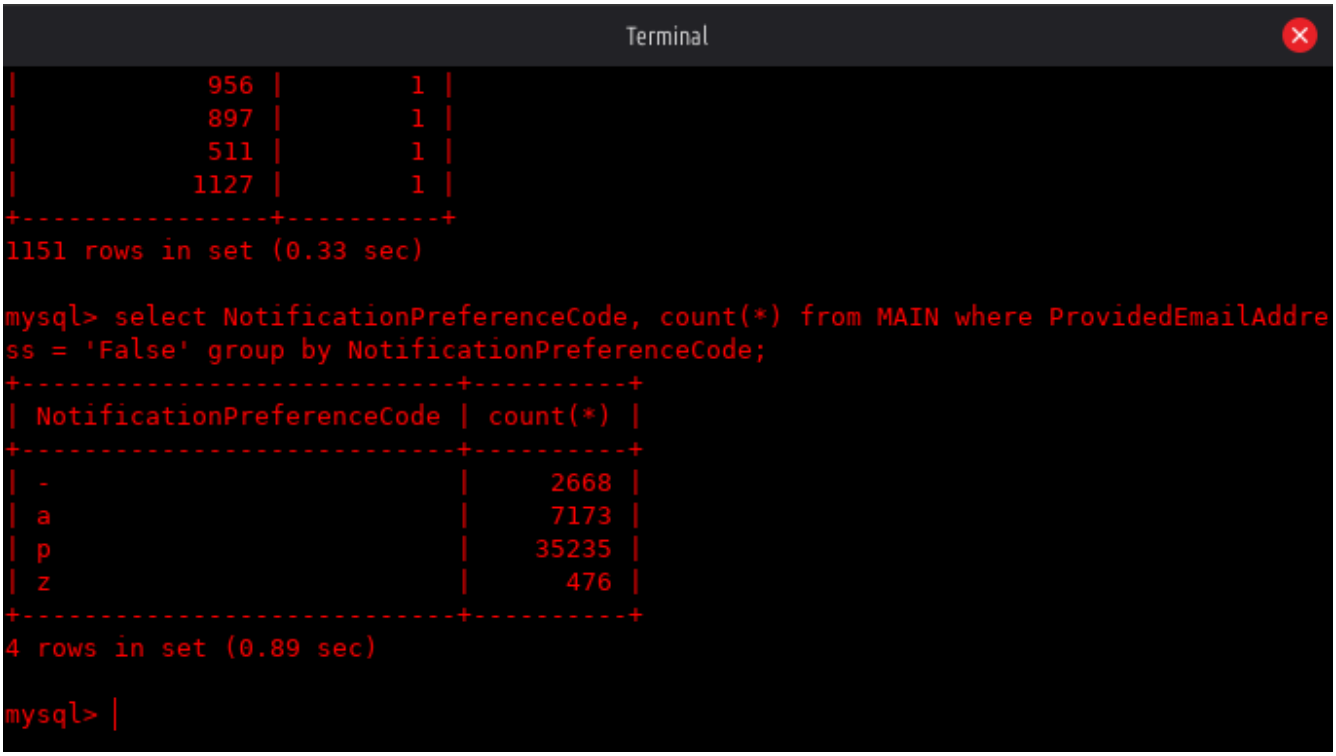
It also appears that most patrons aged 0 to 9 years have not checked out a book.



TotalCheckouts	count(*)
0	14494
53	70
3	747
434	8
18	233
9	369
1687	1
47	75
110	31
279	11
49	74
2	889
756	2
800	2
316	10
43	94
1868	1
658	1
76	55
2071	1
79	46
274	7
935	2
193	22
102	41
126	28
22	197
765	4
19	215
291	23
4	695
52	69
12	276
120	20
153	22
45	89
129	25
368	8
68	62
11	303

This took 0.33 seconds to find and was found using the command “select TotalCheckouts, count(*) from MAIN where AgeRange = '0 to 9 years' group by TotalCheckouts;”.

Lastly, it appears that 476 people have email as their preferred form of contact despite not providing an email address. Most people who did not provide an email address prefer to be contacted by phone.


Terminal
956	1
897	1
511	1
1127	1
+----------------+----------+
1151 rows in set (0.33 sec)
mysql> select NotificationPreferenceCode, count(*) from MAIN where ProvidedEmailAddre
ss = 'False' group by NotificationPreferenceCode;
+---------------------------+----------+
| NotificationPreferenceCode | count(*) |
+---------------------------+----------+
-	2668
a	7173
p	35235
z	476
+---------------------------+----------+
4 rows in set (0.89 sec)
mysql>